# Data Science Capstone - Weekly Report 3

**Project:** MoodBand: An Inclusive Wearable System for Real-Time Mood Detection and Adaptive Sensory Feedback

**Supervisor:** Prof. Rached Zantout

**Student:** Ritik Pravin Mota

**Meeting Info:**

- Date: 5 October, 2026, Monday
- Attendees: Prof. Rached Zantout, Ritik Mota
- Mode: In person

**Last Week Progress / Topics Discussed:**

- Finalized the literature review: confirmed the remaining author details for the iFeeling reference (Shafiq ur Réhman and Li Liu), completing the IEEE-numbered reference list across all 10 sources.
- Shared the completed literature review with Prof. Zantout for review.
- Began revising the full project proposal document to reflect the updated project scope — replacing the outdated custom-hardware sections (hardware design, power supply) with wearable-integration sections, and updating the system architecture diagram, methodology, tools and technologies, budget, and references accordingly.
- Conducted a structured comparison of candidate wearable devices for the sensing layer — Fitbit, Apple Watch, Samsung Galaxy Watch, and WHOOP — evaluated against price, developer API accessibility, real-time data availability, and development environment fit.
- Identified that Fitbit's legacy Web API is being retired by Google (developer support ended September 30, 2026; full shutdown October 30, 2026) and replaced by the new Google Health API, which removes the prior approval bottleneck for intraday heart-rate data.
- Established that the companion app must support both Android and iOS users. This requirement structurally ruled out Apple Watch (pairs with iPhone only) and Samsung Galaxy Watch (continuous real-time streaming is Android-only via the Samsung Health Sensor SDK).
- Finalized on Fitbit, integrated via the Google Health API, as the project's sensing device — the only shortlisted option whose data is not tied to a single phone platform.
- Updated the project proposal's Methodology (Section 7.1) and Tools and Technologies (Section 9) sections to reflect the Google Health API integration.
- Drafted a wearable device comparison document summarizing the evaluation and recommendation, for the supervisor's review and final sign-off on device selection.

**Plan for Next Week:**

- Review the device selection and the updated project proposal with Prof. Zantout and obtain final sign-off.
- Begin designing the Google Health API integration (authentication flow, and the specific data fields to pull — heart rate and motion in real time, with HRV incorporated as a supplementary overnight signal).
- Start planning the mood-induction data collection protocol described in Section 7.2 of the proposal.

**To-Do List:**

- Finalize and share the updated project proposal and device comparison document with the supervisor.
- Set up a Google Cloud / Fitbit developer account and begin exploring the Google Health API documentation.
- Draft the data collection protocol for gathering labelled training data.